

Server-side Languages

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A few generalities

- What are server-side languages ?
 - PHP, ASP, Java, C#, Python, Ruby, etc...
 - Also general purpose languages
- What do they do
 - They generate HTML
 - And much more
 - Connection to DB
 - Writing to files
 - Calling other servers
 - ...

A few generalities

- Frontend and backend
 - Not necessarily on the server
- Usual sensitive points
 - Abstraction (OOP, class, prototype, etc.)
 - Scalability
 - Security (DoS, DDoS...)
 - Complexity (framework/configuration)
 - Performance
 - Concurrent access
 - Cache
 - Sessions

PHP: A very quick introduction

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Definition



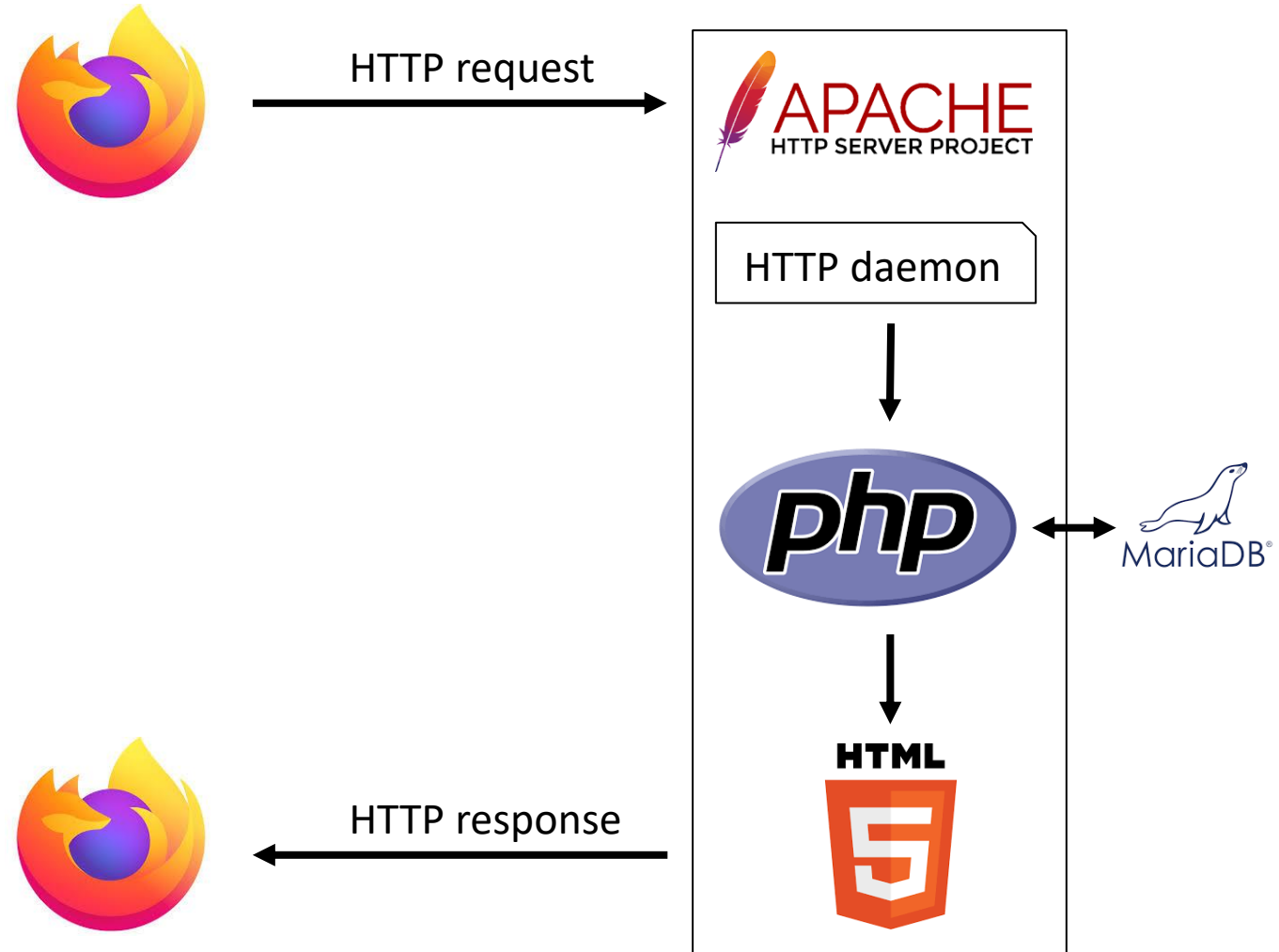
- What is PHP
 - A HTML-embedded, cross platform, server-side scripting language
 - With a recursive acronym (PHP = PHP: Hypertext Preprocessor)
- What does that mean?
 - It is inserted into the HTML code
 - It is interpreted when called, not compiled in advance
 - The interpreter is integrated in the server (Apache module)
 - Runs on all platforms (Microsoft, Linux, Mac...)
 - Generates HTML dynamically

First example

- Hello Word (5 minutes)
- Explain what happens

```
<html>
  <head>
    <title>PHP Test</title>
  </head>
  <body>
    <?php echo '<p>Hello World</p>'; ?>
  </body>
</html>
```

A message exchange overview



Operation

- `<?php` and `?>` forms a PHP block
 - Everything between those tags is interpreted as PHP
 - Everything outside of it is not interpreted
 - It is copied as is
- PHP instructions do not appear in the generated output
- Official documentation
 - <https://www.php.net/manual/en/langref.php>

Operation

- PHP is object oriented
 - More precisely, class-based
 - Syntax is here: <https://www.php.net/manual/en/language.oop5.php>
- PHP supports frameworks
 - Codeigniter, Yii, Symfony, Laravel...
 - Have a look, try them at home

Best PHP frameworks in 2025

Web application framework Software / php / 2025

From sources across the web

	Laravel MIT License	▼		Symfony MIT License	▼		CodeIgniter MIT License	▼
	CakePHP MIT License	▼		Yii BSD licenses	▼		Laminas BSD licenses	▼
	Phalcon BSD licenses	▼		FuelPHP MIT License	▼		PHPixie BSD licenses	▼

1 more ▼

Feedback

Connecting to a DB

- Different ways
 - One obsolete: mysql_* methods -> do not use
 - 2 are okay: PDO and Mysqli
 - <https://www.php.net/manual/en/set.mysqlinfo.php>

Managing passwords

- You want to encrypt them
 - No one, even the admin, should read them
- But...
 - What if many people have the same password ?
- You need a salt
 - Hashing functions are deterministic
 - Salt provides unicity
 - Based on timestamp and IP for example
- <https://www.php.net/manual/en/faq.passwords.php>